**CSA1739-ARTIFICIAL INTELLIGENCE**

**PRALOG PROGRAMS**

**RAHIMA BANU N**

**192525009**

**DAY 2:**

7. Sum of Integers from 1 to N

```
4 ?- consult(['c:/Users/rahim/Downloads/integers.pl']).
true.

5 ?- sum(5,S).
S = 15 .
```

8. Database with Name and DOB

```
12 ?- consult(['c:/Users/rahim/Downloads/dob.pl']).
true.

13 ?- show_dob('arun',DOB).
DOB = '2003-05-12'.

14 ?- person(Name,DOB).
Name = arun,
DOB = '2003-05-12'
```

9. Student-Teacher-Subject Code Database

```
15 ?- consult(['c:/Users/rahim/Downloads/student.pl']).
true.

16 ?- show_details('arun',Teacher,Subject).
Teacher = raja,
Subject = 'CS101'.

17 ?- student_teacher(Student,Teacher,Subject).
Student = arun,
Teacher = raja,
Subject = 'CS101'
```

10. Planets Database

```
18 ?- consult(['c:/Users/rahim/Downloads/planets.pl']).
true.

19 ?- show_planet(earth,N).
N = 3.

20 ?- planet(Name,Number).
Name = mercury,
Number = 1
```

**DAY 3-**

1. Towers of Hanoi

```
23 ?- consult(['c:/Users/rahim/Downloads/hanoi.pl']).
true.

24 ?- hanoi(3,a,c,b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
```

2. Bird Flying

```
25 ?- consult(['c:/Users/rahim/Downloads/bird.pl']).
true.

26 ?- check(parrot).
parrot can fly.
true .

27 ?- check(penguin).
penguin cannot fly.
true.

28 ?- check(sparrow).
sparrow can fly.
true
```

3. Family Tree

```
29 ?- consult(['c:/Users/rahim/Downloads/male.pl']).
true.

30 ?- father(ram,raj).
true.

31 ?- mother(sita,raj).
false.

32 ?- sibling(raj,sita).
true.
```

4. Diet System

```
33 ?- consult(['c:/Users/rahim/Downloads/diet.pl']).
true.

34 ?- suggest(diabetes).
Suggested diet: Low sugar food
true.
```

5. Monkey Banana

```
35 ?- consult(['c:/Users/rahim/Downloads/move.pl']).
true.

36 ?- can_get_banana.
Monkey gets the banana
true.
```

6. Fruit and Color

```
37 ?- consult(['c:/Users/rahim/Downloads/fruit.pl']).
true.

38 ?- fruit_color(X,Y).
X = apple,
Y = red ;
X = banana,
Y = yellow ;
X = grapes,
Y = green ;
X = Y, Y = orange.
```

7. Best First Search

```
39 ?- consult(['c:/Users/rahim/Downloads/edge(a, b, 1).pl']).
true.

40 ?- bestfirst(a,d).
Goal Found: d
true .
```

8. Medical Diagnosis

```
41 ?- consult(['c:/Users/rahim/Downloads/symptom(ravi, fever).pl']).
true.

42 ?- diagnose(ravi).
Disease: flu
true .

43 ?- diagnose(anu).
Disease: headache_problem
true.

44 ?- diagnose(ram).
false.
```

9. Forward Chaining

```
45 ?- consult(['c:/Users/rahim/Downloads/forward.pl']).
true.

46 ?- rule1.
Ram is mortal
true.
```

10. Backward Chaining

```
47 ?- consult(['c:/Users/rahim/Downloads/human(ram).pl']).
true.

48 ?- mortal(ram).
true.
```